

SI-CMP-032 - Compressor System Safety Instructions

Document ID: SI-CMP-032 Revision: 1.9 Effective Date: 1 October 2025 Site: EastRefinery Assets: Compressor C-201, Compressor C-202 Owner: Process Safety / Area Operations

1. Key hazards

1. High-pressure gas release from flanges, coolers, and instrument points
2. Mechanical damage and instability during surge
3. High discharge temperature and ignition risk on flammable services
4. High noise and vibration
5. Unexpected movement of the recycle / anti-surge valve
6. Hot cooler surfaces and fan hazards

2. Rules during discharge pressure alarms

1. Check wind direction before any field approach.
2. Do not stand in line with flanges, cooler headers, or known leak points.
3. Keep anti-surge protection in service.
4. If discharge temperature is also high, prioritise cooling and load reduction over keeping rate.
5. On a confirmed major leak, evacuate and start emergency response.
6. Tell the board when you enter and leave the compressor area.

3. Prohibited actions

1. Inhibiting anti-surge because the alarm list is noisy
2. Forcing recycle closed while discharge pressure is already high
3. Entering cooler bays during unstable compressor operation without authorisation and a clear need
4. Resetting trips and restarting without understanding the cause

4. Field approach judgement

Ask before walking in:

- Is the machine surging or swinging hard?
- Is there a smell, noise change, or visible leak?
- Can the board do the next useful action remotely?

If the useful next action is board-side, stay out of the area.

5. Conflict with advisory recommendations

If Alarm API or copilot text conflicts with this safety instruction, follow SI-CMP-032 and escalate. Safety instructions win over advisory recommendations.

6. Related documents

- OP-CMP-002

- MM-CMP-011
- TG-CMP-021
- SI-GEN-030